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August 11, 1959

FEDERAL SPECIFICATION

DISHWASHING MACHINE, HOUSEHOLD, (ELECTRIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers electric household dishwashing machines, commercially produced for use in domestic kitchens for under-counter installation.

1.1.1 *Federal specification coverage.* — Federal specifications do not cover all types, classes and styles of the commodity indicated by the title of the specification, but include only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types.*—Dishwashing machines covered by this specification shall be of one capacity and of the following types, as specified (see 6.2):

Type I.—Gravity drain.

Type II.—Pump drain.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

QQ-S-766 — Steel Plates, Sheets, and Strip, Corrosion-Resisting.

UU-P-271 — Paper, Wrapping, Waterproofed Kraft.

UU-T-111 — Tape; Paper, Gummed (Sealing and Securing).

PPP-B-591—Boxes, Fiberboard, Wood-Cleated.

PPP-B-601—Boxes, Wood, Cleated-Plywood.

PPP-B-636—Boxes; Fiber.

PPP-T-60 — Tape; Pressure Sensitive Adhesive, Waterproof—for Packaging and Sealing.

PPP-T-97 — Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver,

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San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–P–116—Preservation, Methods of.

MIL–B–117—Bags, Interior Packaging.

MIL–B–121—Barrier Material, Greaseproofed, Flexible (Waterproofed).

MIL–A–140 — Adhesive, Water-Resistant, Waterproof Barrier-Material.

MIL–V–173 — Varnish, Moisture- and Fungus-Resistant, for the Treatment of Communications, Electronic, and Associated Electrical Equipment.

MIL–V–1137 — Varnish, Electrical-Insulating (For Electro-Motive Equipment).

MIL–C–10382 — Corrosion-Preventive, Petrolatum, Spraying Application; for Food Handling Machinery and Equipment.

MIL–C–11133 — Crates; Wood, Slatted Style, Wirebound, Domestic.

MIL–C–11456 — Crates; Wood, Nailed, Unsheathed; Domestic Shipment, 1500 Pounds Maximum Load.

MIL–I–16910 — Interference Measurement, Radio, Method and Limits; 14 Kilocycles to 1000 Megacycles.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–109 — Inspection Terms and Definitions.

MIL–STD–129—Marking for Shipment and Storage.

MIL–STD–130 — Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Nongovernmental:

Underwriters' Laboratories, Inc. Standards.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York 13, N. Y.; 207 East Ohio Street, Chicago 11, Ill.; or 1655 Scott Lane, Santa Clara, Calif.)

National Electrical Manufacturers' Association Standards:

HS–1—Place Setting for Determining Capacity of a Household Electric Dishwasher.

MG–1—Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 155 East 44th Street, New York 17, New York.)

Uniform Freight Classification:

Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

3. REQUIREMENTS

3.1 Preproduction sample.—For the Department of Defense purchases and for others when specified (see 6.2), before production is commenced, a sample of the finished dishwashing machine to be furnished shall be made available to the contracting officer for approval. The sample may be included in the quantity required for the final shipment. Approval of sample by the contracting officer shall not obligate the Government to accept equipment which is not in accordance with this specification, nor shall it relieve the contractor of obligation to supply equipment specified herein. Any work done on the contract prior to approval of the preproduction sample will be at the contractor's own risk.

3.2 Standard product. — Each domestic dishwashing machine shall be a current standard commercial product except for such

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deviation from manufacturer's model as may be required by the detailed requirements herein. Component parts of the dishwasher need not be the product of the same manufacturer.

3.3 Certification.

3.3.1 *Contractor's certificate.*—When applicable, in a letter notifying the contracting officer that the preproduction sample is ready for inspection and test, the contractor shall certify that satisfactory preliminary operating and performance tests of the preproduction sample have been made and that this item has been manufactured or assembled under the same management and the same or similar manufacturing methods and facilities as will be employed on the production items. Evidence of accepted purchase orders for components, fabrication or assembly by subcontractors and suppliers, in sufficient detail to demonstrate compliance with this specification and inspection of preliminary test data, may be requested by the contracting officer.

3.3.2 *Approval requirements.*—The bidder shall submit proof satisfactory to the contracting officer, with bid, that the equipment which he proposes to furnish conforms to requirements of the Underwriters' Laboratories, Inc. regarding fire and casualty hazards, and National Electrical Manufacturers' Association Standards for electrical controls and motors. The label or listings of Underwriters' Laboratories, Inc., as applicable, will be accepted as conforming to this requirement.

3.3.3 *Proof of approval requirements.*—In lieu of the label or listing of the Underwriters' Laboratories, Inc., the bidder may submit, for approval of the contracting officer, certified test reports of a national recognized independent laboratory competent to perform such test indicating conformance to all requirements of the applicable Underwriters' Laboratories, Inc. Standard, including method of test.

3.4 Materials.—The material specified shall conform to the applicable specifications and requirements specified hereinafter. The contractor shall be required to perform the tests for compliance with the characteristics specified in 4.4.1.2. Compliance with the characteristics of 4.4.1.2 in no way relieves the contractor from furnishing materials which comply with other requirements of the referenced specifications or other requirements specified herein or precludes the Government from performing such further tests as provided for in this specification or referenced specifications.

3.4.1 *Steel.*—Steel used in the fabrication of the tub shall be commercial grade 1010 or better and have a minimum nominal wall thickness of 0.0299 inch. Frame, braces and other parts shall be commercial heavy-duty steel.

3.4.2 *Corrosion-resistant steel.*—Corrosion-resistant steel shall conform to classes 202, 302, 304, 316 or 430 of Federal Specification QQ–S–766. Welding for class 430 corrosion-resistant steel shall be done with commercial 308, 309 or 310 welding rods.

3.4.3 *Plastic or rubber compound for racks.*—Racks for dishes and silverware shall be made of corrosion-resistant material and covered with a plastic or rubber compound (see 3.6.12).

3.4.4 *Motors.* — Motors shall conform to NEMA Standard MG-I, Standards for Motors and Generators. Motors for recirculation or washing solutions and rinses shall be not less than ⅙ horsepower.

3.4.5 *Gaskets.*—Gaskets exposed to washing solutions shall be fabricated of material which does not contain sulfur and which will resist deterioration by heat or detergents. Manufacturer's certificate of compliance will be acceptable.

3.4.6 *Rubber tubing.*—Rubber tubing shall be of the type normally used by the manufacturer on dishwashing machines.

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3.4.7 *Piping, fittings, and valves.*—Piping, fittings, and valves for the control of water, drains, and overflows shall be made of corrosion-resistant material.

3.4.8 *Hardware.* — All bolts, nuts, pins, screws, and other fastenings or fittings shall be standard commercial quality. Those coming in contact with water solution shall be corrosion resistant.

3.4.9 *Varnish sealer.*—Varnish sealer shall conform to type M, grade BB or CB of Military Specification MIL–V–1137.

3.4.10 *Varnish fungus resistance.*—Fungus resistance, when specified (see 6.2), shall conform to Military Specification MIL–V–173.

3.5 **Design.**—The dishwasher shall be designed for under-counter installation with either a pull-down door with slide-out racks or a pull-out drawer style. Machine shall also be designed so that during wash and rinse cycles, water shall be circulated through the load for such a time as to wash and rinse all pieces throughout the rack area. The water distribution shall be such that direct or reflected streams of water shall contact all interior and exterior surfaces of the dishes and utensils to be washed.

3.5.1 *Component parts.*—The dishwashing machine shall be provided with the following major components: Racks, strainer, spraying system, and detergent dispenser.

3.6 **Construction.**—Dishwashing machines shall be rigid and rugged, free of objectionable vibration, quiet in operation, when tested as specified in 4.5.1, and all parts requiring adjustment shall be readily accessible.

3.6.1 *Dimensions.*—Overall exterior dimensions shall not exceed the following:

	Inches
Width	24 – 26
Depth	24 – 26
Height	34 – 36

3.6.2 *Electrical requirements.* — Machine shall operate on 115 volts, 60 cycles, single phase, alternating current.

3.6.2.1 *Automatic control.*—The automatic control shall be designed to cycle the machine through the wash, rinse, dry or heat, and stop phases. Also, the automatic control shall be capable of manual setting to repeat or skip any phase of the cycle.

3.6.2.2 *Switch safety-interlock.* — Machine shall be equipped with a safety interlock switch, which shall automatically stop the spraying action in the tub when the door or drawer is opened when tested as specified in 4.5.1.

3.6.2.3 *Heater, electric.*—The heater shall be the encased type having the resistance wire encased in a sealed metal tubing or sheathing and have a rated capacity of not less than 600 watts input for drying and for warming when tested as specified in 4.5.1.

3.6.2.4 *Motors.* — Determination shall be made that motors specified in 3.4.4, are of the proper type required, have proper speed and ample power and drive mechanism shall be checked for proper alignment and the motors shall show no undue vibration, when tested as specified in 4.5.1.

3.6.2.5 *Fungus and moisture resistance (see 6.4).* — When specified (see 6.2), the equipment shall be treated to resist fungus and moisture as follows:

3.6.2.5.1 Circuit elements such as motor coils, generator and transformer windings, and similar electrical components, which have a temperature rise exceeding 75°F. when operating at full load, shall not be coated with a fungitoxic compound. Instead, such components shall be given two coats of varnish conforming to type M, grade BB or CB and one sealer coat conforming to type M, grade CB specified in 3.4.9. The coats shall be applied by the vacuum-pressure, immersion, centrifugal, pulsating pressure or built-up method

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so as to fill all interstices in the coils and preclude the entrapment of air or moisture. The sealer coat may also be applied by brushing or spraying.

3.6.2.5.2 Electrical components such as switches, fuses, contacts, and heater elements, shall not be treated. Other material and components which are inherently fungus resistant or are protected by hermetic sealing need not be treated.

3.6.2.5.3 Circuit elements, not covered above and which have a temperature rise of not more than 75° F. when operating at full load, shall be coated with fungus-resisting varnish as specified in 3.4.10. Circuit elements include, but are not limited to cable, wire, switch boards, panel boards, terminal and junction blocks, junction boxes, and coils and capacitors.

3.6.3 *Spray assembly.*—The spray mechanism shall be designed for easy removal and shall be easy to clean in place or when removed.

3.6.4 *Strainer (scrap trays).*—Scrap trays or strainers shall be corrosion-resistant and shall be provided as necessary to effectively prevent insoluble waste matter affecting the operation of the pump, and large pieces of food residue from passing into the discharge system. There shall be no opening around or between the trays wider than the diameter of the openings in the trays. The design and construction of the trays and the ledges on which they rest shall be such that the trays are easily removable from the front of the machine for cleaning and, when removed, all the surfaces beneath the ledges are easily accessible for cleaning and resistant to recommended detergents.

3.6.5 *Pumps.*—When machine is furnished with a pump drain, two pumps may be provided if necessary for proper operation, otherwise a single pump may be used. When gravity-drain type machines are furnished, only one pump is required. Pumps shall be of the centrifugal type and so designed that normal particles of food waste in the drain water will not affect its operation. The pumps shall be of the type normally furnished by the manufacturer on this kind of equipment. The water-handling system shall be able to resist attack by the recommended dishwashing detergents. Manufacturer's certificate of compliance to resist attack will be acceptable.

3.6.5.1 *Bearings.*—Pump motors, or pump and motor units shall be provided with suitable bearings designed for long life.

3.6.5.2 *Seals.* — Pumps shall be equipped with rotary mechanical shaft seals of the self-lubricating type.

3.6.6 *Lubrication.*—All bearings of pumps, motors and rotating parts shall be adequately lubricated for the life of the equipment.

3.6.7 *Piping, fittings and valves.* — Piping, fittings and valves specified in 3.4.7 shall be installed on the dishwashing machine by the contractor, and shall be so arranged that a minimum of pipe fitting shall be necessary.

3.6.8 *Water supply.*—The incoming water system for the machine shall be capable of carrying hot potable water without contamination and to withstand a maximum temperature of 160° F. The water supply to the dishwasher shall have either an air gap of one inch or twice the internal inlet diameter, whichever is greater or a vacuum breaker. Machine shall be capable of satisfactory operation on flow pressure of 16 to 120 pounds per square inch, when tested as specified in 4.5.1.

3.6.9 *Drain.*—The machine shall be either gravity drain or pump drain as specified. Machines supplied with a pump drain shall be equipped with a drain air gap either as an integral part of the machine or as an accessory.

3.6.10 *Tub.*—The tub shall be fabricated of steel specified in 3.4.1 and so designed as to provide a rapid drawoff of water.

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3.6.11 *Gaskets.*—The gaskets, fabricated of material specified in 3.4.5 shall be so secured that the gasket will shed water in such a manner that putrescible matter will not be entrapped.

3.6.12 *Racks.*—Racks covered with plastic or rubber compound specified in 3.4.3 shall not deteriorate, warp or alter in shape under normal usage. Manufacturer's certificate of compliance will be acceptable.

3.6.12.1 *Loading of dishwashing machine.*—Racks shall be designed to prevent nesting and flop-over of dishes and to assure uniform washing of each dish, when tested as specified in 4.5.1.

3.6.13 *Detergent control.* — The machine shall be provided with a detergent dispenser which shall dispense detergent during operation of the machine and shall be of a type generally supplied by the manufacturer.

3.6.14 *Pull-down door and pull-out drawer dishwashing machines.*—Either of the following may be furnished:

3.6.14.1 *Pull-down door.*—Pull-down door dishwashing machine with slide-out racks shall be constructed in such a manner as to prevent water leakage when in a closed position, and shall be capable when open of supporting the lower rack filled with dishes, without permanent distortion when tested as specified in 4.5.1. Door shall be counter-balanced and the machine so constructed as to cushion adequately the force of the impact when the door is opened to the horizontal position. When the door is open with racks fully extended and completely loaded, the machine shall exhibit no tendency to tip forward when permanently installed in accordance with installation instruction.

3.6.14.2 *Pull-out drawer.* — The pull-out drawer dishwashing machine shall be constructed in such a manner as to prevent leakage of water when in a closed position and drawer shall slide easily without binding, when tested as specified in 4.5.1. The machine shall be constructed in such a manner that there shall be no tendency to tip forward when permanently installed in accordance with installation instructions. A stop shall be provided to keep the drawer or drawers from falling out when fully extended, and shall permit removal of the drawers when necessary. Drawer should slide easily without binding and racks should be readily removable for cleaning.

3.6.15 *Interchangeability.* — All component parts of a completed dishwashing machine shall be interchangeable with parts of other machines furnished under this specification on any one contract.

3.6.16 *Metal fabrication.*—Sheet metal or structural sections used in the fabrication of dishwashing machines shall be free of kinks, bends and winds (except kinks intentionally used for design or strength). All internal seams resulting from the joining of two or more metal edges in the construction of tubs or other parts exposed to wash or rinse water, shall be made so that the finished seam is completely closed and the surface is smooth. Bolt and rivet holes required for mounting compounds shall be punched or drilled and shall have all burrs removed. Rivets shall be driven to completely fill the holes, with full sized heads neatly finished and in full contact with the surface of the member. Nuts and screws shall be drawn tight.

3.6.17 *Welding.*—All welds shall be sound and smooth, free from pits, holes or fissures. The deposited weld metal shall be thoroughly fused with the base metal along all surfaces and edges of union and shall penetrate into the root of the joint. Spotwelds shall be free from pits and flashes.

3.6.18 *Radio interference suppression (see 6.4).*—When specified (see 6.2 and 6.3), dishwashing machines procured under this specification shall be equipped for suppression of

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radio interference in accordance with Military Specification MIL–I–16910.

3.7 Performance characteristics.—The machine shall be capable of washing eight standard place settings as defined in NEMA Standard, HS–1, and eating utensils physically and hygienically clean. The machine shall be automatically controlled to permit a wash, rinse, dry and turn-off phase. The time required for the complete cycle shall not exceed a total of 60 minutes with water supply temperature of not less than 140°F. Water required for the complete cycle shall not exceed thirteen gallons.

3.8 Finish. — Unless otherwise specified (see 6.2), the front panel or door front, and sides (if any) shall be finished with white baked enamel. The interior of the steel tub specified in 3.6.10 shall be covered with a porcelain enamel or bonded plastic finish which is acid and alkali resistant. All other metal surfaces shall be corrosion resistant, painted or plated.

3.9 Operation, installation and maintenance instruction.—Instruction booklets giving operation, installation, and maintenance instructions (including pictures as necessary), dimensions, floor plan of machine, electrical data, and connection diagrams for all utilities shall be furnished by the contractor and packed with each machine. Instructions for leveling the dishwasher shall be included in the installation instructions. For Department of Defense purchases and when specified, instructions for each machine shall also contain a complete parts list with each part properly identified for exact duplication.

3.10 Marking for identification.

3.10.1 *Civil agencies.* — Unless otherwise specified (see 6.2), the manufacturer's standard nameplate will be acceptable.

3.10.2 *Military agencies.* — Identification marking shall be on a corrosion-resistant metal plate, and in accordance with Military Standard MIL–STD–130, with the following information: Nomenclature, stock number, serial number (if applicable), manufacturer's identification, part or model number, date of manufacture, contract number, the notation "U.S." and any special characteristics. In addition, electrical characteristics shall appear on a plate.

3.11 Workmanship. — The finished article shall be clean, well made, and free from defects which may affect appearance or serviceability.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Inspection responsibilities. — Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and made available to the Government as specified in the contract or order.

4.2 Preproduction inspection.

4.2.1 *Preproduction sample inspection.*— When preproduction inspection is specified, examination and testing of the preproduction sample shall be made on a completely fabricated dishwashing machine for all provisions of this specification applicable to end product examination and tests (see 4.4.3.1.1).

4.3 Sampling for inspection and acceptance. —Inspection shall be made in accordance with the provisions set forth in Military Standard MIL–STD–105, except where otherwise indicated hereinafter. Inspection terms used in this inspection shall be interpreted in accordance with the definitions in Military Standard MIL–STD–109.

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4.4 Examination.

4.4.1 *Inspection of components and materials.*

4.4.1.1 *Visual and dimensional examination.*—Inspection shall be made of components and materials to determine compliance with the visual and dimensional requirements of this specification and applicable referenced specifications provided that the determinations cannot be made during examination of the end item. This inspection shall include determination for compliance of finish, type, size, grade, thickness, and class, as specified for the various components or materials. Any deviation from specified requirements shall be classified as a defect. The inspection level shall be L–2 with an acceptable quality level of 4.0 defects per one hundred units.

4.4.1.2 *Testing of components and materials.*—Tests shall be performed on representative samples from each lot or batch of materials used in the fabrication, packaging and packing of the end item to determine compliance with this specification and with applicable specifications as hereinafter required. The supplier shall furnish certified test results for the component characteristics listed in table I. The supplier shall furnish certified test results for the component characteristics listed in table I. The supplier shall furnish a separate certificate for the characteristics for each different component not listed below, certifying that the components comply in all particulars with the materials specified in sections 3 and 5 of this specification. The Government reserves the right, at its option, to verify the correctness of such certificates.

4.4.2 *Intermediate inspection.*

4.4.2.1 *Visual and dimensional examination.*—The dishwashing machine shall be examined for compliance with all internal visual and dimensional requirements of the end item. Examination shall be made at appropriate intermediate stations. Assemblies will be examined for defects in construction, workmanship, dimensions and materials. Any deviation from specified requirements shall be classified as a defect. The inspection level shall be L–2 with an acceptable quality level of 4.0 defects per 100 units.

4.4.3 *End product inspection.*

4.4.3.1 *Examination of the end product.*—The defects found during examination shall be classified in accordance with 4.4.3.1.1 and 4.4.3.1.2. The sample unit for these examinations shall be one complete dishwashing machine.

TABLE I.

Component	Characteristics	Specification reference	Sample unit size	Acceptance number
Steel, CRES	Chrome, nickel, and carbon content	QQ–S–766	¼ ounce piece 3	0
Welding rod[1]	Chrome, nickel, and carbon content	Basic	4 ounce 1	0
Fiberboard[2]	Bursting strength	PPP–B–636	See PPP–C–636	
Barrier material	Greaseproofness, waterproofness (as received)	MIL–B–121	1 square yard 1	0

[1] Kind of rod tested shall be identified on test reports (i.e. 308, 309 or 310).
[2] Box maker's certificate may be accepted in lieu of verification testing.

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4.4.3.1.1 *Examination of the end product.*—Examination shall be made of representative samples of the completely assembled end item. Examination for defects in the following classification is based upon the assumption that these defects, if they do occur, will occur on a random basis. (However, certain nonrandom type defects have also been included, where no inspection time is saved by grouping them separately). The inspection level shall be level II and the acceptable quality level (AQL) shall be 2.5 for major defects and 6.5 for total defects expressed in defects per 100 units.

TABLE II.—*Classification of defects*

Examine	Defect	Classification		
		Major	Minor	Major[1] or Minor[2]
Finish	Not finished (ferrous metal)	X		
	Type of finish not as specified, or blistered, peeled, chipped, not adherent, or area of no film or thin film		X	
	Poor touchup, color separation, scratched, stained, overspray, imbedded foreign matter or otherwise impaired		X	
	Area of rust		X	
	Texture of metal finish not as specified or exposed machine marks affecting appearance		X	
Materials	Any component not fabricated of the specified material (except as otherwise indicated herein)	X		
	Fungus proofing, when required, not as specified	X		
Design	Any characteristic that is not in accordance with the specified requirements. (Except as otherwise indicated)	X		
	Any component not interchangeable with like part on any unit manufactured under the same contract	X		
	Verify that an air gap or vacuum breaker is provided	X		
Type	Not type specified	X		
Construction and workmanship (General) (Applicable to all components and assemblies)	Any part missing or not specified type or size (except as otherwise indicated herein)	X		
	Fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, malformed, decayed, or otherwise impaired			X
	Exposed belts, chains, shafts, gears, et cetera, not fully enclosed or guarded	X		
	Guard, et cetera, not rigid and readily removable without disassembling pipes or fittings	X		
	Misplaced, loose, or not in proper alignment			X
	Sharp burr, sliver, or splinter that may be injurious to personnel but does not affect the performance of the unit			X
	Any operation omitted or not properly performed			X
	Any functioning component that is inoperative or will not operate as intended	X		

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TABLE II.—(Continued)

Examine	Defect	Classification		
		Major	Minor	Major[1] or Minor[2]
Construction and workmanship (General) (Applicable to all components and assemblies) (cont'd)	Any functioning component that required abnormal force to be operated	X		
	Not connected or joined as specified, or poorly accomplished, seriously affecting serviceability	X		
	Control, valve, switch, et cetera, not mounted in a manner that affords the maximum protection against accidental damage		X	
	Any adjustable assembly that cannot be adjusted, or is not properly adjusted to perform the function intended	X		
	Component not readily accessible for servicing where required		X	
Welding	Slag inclusion, undercut, not smooth and uniform, scale or flux deposit not removed			X
	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused.	X		
Bolts, nuts, screws, studs, and other types of threaded components	Missing, not specified type, broken, stripped, fractured, or loose			X
	Lock washer or washer missing where required		X	
Rivets	Missing, not specified type, loose, broken, not peened, or insufficiently peened			X
Electrical apparatus wiring, et cetera	Wiring cut, abraded, not properly joined or loose at terminal seriously affecting serviceability	X		
	Not enclosed in specified conduit when required	X		
	Not watertight, where required	X		
	Wiring loose at terminal, adequate slack not provided for wiring to relieve strain, excessive insulation stripped from wiring, strain relief device or support clip missing	X		
Controls, valves, gages, nozzles, pipe, drains, et cetera	Not properly calibrated, not specified capacity or type, or otherwise defective affecting use	X		
	Improperly connected or installed, will not provide a seal where required, stripped, creased, restricting or partially restricting interior flow seriously affecting seviceability	X		
	Defective but not seriously affecting serviceability		X	
Cleanliness of unit	Not clean, but not seriously affecting serviceability		X	
Lubrication	Lubrication not provided	X		
	Lifetime lubrication not provided	X		
Assembly	Component not properly assembled, affecting serviceability, e.g., motor or pump coupling out of alignment	X		
	Component not properly assembled or secured but not seriously affecting serviceability, e.g., lock washer missing from fastener		X	

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TABLE II.—(Continued)

Examine	Defect	Classification		
		Major	Minor	Major[1] or Minor[2]
Assembly (cont'd)	Unit or sub-assembly perceptibly out of square or alignment	X		
	Access panel, hood or door not properly hung, warped, sprung, or not properly counter-balanced where required	X		
Marking (Instructions and identification)	Missing, incomplete, not legible, not specified type, or size, not in proper location, not affixed in the prescribed manner, not correct color; plate label, or tag not fabricated of the specified material, not stenciled, stamped, embossed, printed, lithographed, et cetera, whichever required.		X	

[1] Major if seriously affecting serviceability.
[2] Minor if not seriously affecting serviceability.

4.4.3.1.2 *Dimensional examination.*—Representative samples of dishwashing machines shall be examined to determine compliance with the dimensional requirements specified herein. Any dimension not within the specified tolerance shall be classified a defect. The inspection level shall be L–2 and the acceptable quality level shall be 6.5 defects per 100 units. Requirements are applicable to the individual unit.

4.4.3.2 *Testing of the end product.*—The preproduction sample shall be tested by the contractor as provided in 4.2.1. Testing of the preproduction sample will be observed by a Government representative. The functional test specified in 4.5.1 need not be repeated on production units furnished unless there is a change in design material or method of construction. Each dishwashing machine shall be tested as specified in 4.5.2.

4.4.3.3 *Inspection of preparation for delivery.*—Examination will be made of preservation, packaging, packing and marking to determine compliance with requirements of section 5 of this specification. Units and crates shall be specifically inspected for compliance with the characteristics listed below:

a. Draining of units.

b. Lubrication.

c. Bolting, blocking and bracing and nailing practice.

d. Design of crate and size of members.

e. Clearance and marking.

Any deviation from specified requirements shall be classified as a defect. The inspection level shall be L–7 with an acceptable quality level of 6.5 defects per one hundred units.

4.4.3.4 *Reinspection and retest.*—Rejected machines may be reworked and represented for inspection provided appropriate action has been taken to correct the reported deficiencies.

4.4.3.5 *Code and standard requirements.*—The contractor shall furnish the Government inspector with proof of compliance with 3.3.2, 3.3.3 and 3.4.4.

4.5 Tests.

4.5.1 *Functional test.* — Unless otherwise specified (see 6.2), a completely assembled dishwashing machine shall be connected and run through as many cycles as shall be necessary to determine the following:

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1. With the inlet water pressure regulated at 16 to 120 p.s.i., with the machine loaded to capacity with soiled eating utensils (NEMA Standard HS–1) and using detergents recommended by the manufacturers, verify that the stream of water from the spray jets contact and wash all interior and exterior surfaces of the dishes and utensils.
2. That the unit cycles properly and within the required time limits and that the electrical connections are correct.
3. That the cycles can be repeated or skipped and that the safety switch operated.
4. That the water consumption is within specified limits.
5. Each dishwasher with switch on "dry" position shall be given a dielectric strength test to indicate that the heater is capable of withstanding without breakdown for a minimum period of one minute, the application of 60 cycles alternating potential of 900 volts. The dielectric test shall also be repeated for the motor and associated wiring of the unit.
6. That no water leakage occurs.
7. That the required pump(s) function properly.
8. There shall be no undue noise or vibration.
9. Each moving part of the assembled dishwasher shall be checked for smoothness of operation and fit of all controls and moving parts. The dishwasher may or may not be connected to a power source for this determination.

4.5.2 *Operational test.*—Unless otherwise specified (see 6.2), the assembled dishwashing machine shall be connected simulating the actual installed condition and operated for one complete cycle. Determination shall be made that all operational functioning comply with specified performance and established design.

4.5.3 *Radio interference suppression.*—When radio interference suppression is required, it shall be in conformance with Military Specification MIL–I–16910, the preproduction sample shall be tested as described in that specification with Government test equipment by authorized representatives of the contracting officer. Upon approval of the sample by the contracting officer, and provided all other requirements of this specification are met, the test report shall be used as the basis for determining conformance of production units with visual inspection requirements governing the suppression system.

5. PREPARATION FOR DELIVERY

(Civil agencies.—Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 **Preservation and packaging.**—Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Cleaning.*—All exposed, uncoated, unpainted, ferrous metal surfaces of the dishwasher and components shall be cleaned process C–1 of Military Specification MIL–P–116 and thoroughly dried prior to application of preservative.

5.1.1.2 *Repainting and recoating.*—Painted and permanently coated surfaces from which the paint or permanent coating has been removed or which are defective shall be repainted or recoated with material identical with the original.

5.1.1.3 *Preservation.* — All cleaned, uncoated, unpainted, ferrous metal surfaces shall be coated with preservative conforming to Military Specification MIL–C–10382. All exposed surfaces of the dishwasher and component parts, coated with preservatives that might contact other surfaces, shall be covered

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or wrapped with grease-proof, barrier-material conforming to type II, grade A of Military Specification MIL–B–121. Covering or wrapping shall be secured with tape conforming to Federal Specification UU–T–111 or PPP–T–60, or jute, sisal, Manila or cotton twine, or heat-sealed when heat-sealable barrier material is used.

5.1.1.4 *Packaging.* — Detached component parts, such as spray assemblies, filter assemblies, knobs, dials, et cetera and installation hardware, tubing and accessories, when supplied, shall be packaged method III of Military Specification MIL–P–116. The instruction manual and parts list shall be placed in a bag conforming to type III, class b of Military Specification MIL–B–117 and the bag sealed. The vane pump, when supplied, shall be provided with a protective covering. The silverware basket, china rack, and the detergent dispenser, when supplied, shall be secured to one of the dish racks. The lower dish rack shall be secured in place or on the bottom of the washing compartment. The upper dish rack shall be secured in place or to the bottom rack. Racks on tracks or reversible roll racks in place shall have cushioning material or nontest wax-coated corrugated-fiberboard cut pads, strips or trays placed between the rollers and the track and ends of the rollers and sides of the dishwasher so as to preclude damage to the interior of the dishwasher or the flattening of the rollers. The packaged detached components, accessories, and instruction manual that can fit into the dishwasher and the suspended or loose interior components shall be cushioned, blocked, and secured within the dishwasher in a manner to prevent movement or damage while in transit or storage. Pull-down doors shall have a corrugated pad, waxed both sides, placed between the door and racks. The door shall be blocked or cushioned ajar (slightly open) so as to preclude setting or sticking of the water-tight door seal. The door shall then be secured to preclude movement with minimum 1½-inch-width tape conforming to type III of Federal Specification PPP–T–97 extending around the dishwasher and overlapping itself not less than six inches. Top loading drawer-type dishwashers shall have the drawer immobilized in the ajar (slightly open) or latched position by J-bolts, fiberboard pads, cushioning material, or any combination of these materials. When other than J-bolts are used for securing, the drawer shall be secured with tape as specified for the pull-down door. Installed handles, knobs, and dials shall be protected with cushioning material secured with tape conforming to type II or III, class 1 of Federal Specification PPP–T–60.

5.1.1.5 *Caution tag.* — A red caution tag giving explicit deprocessing instructions shall be attached to each dishwasher in a conspicuous location. These instructions shall include a warning to the effect that all tape must be removed and blocked components released prior to operating.

5.1.2 *Level C.*—Each dishwasher shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 **Packing.**—Packing shall be level A, B or C as specified (see 6.2).

5.2.1 *Level A.* — Each dishwasher, preserved and packaged as specified in 5.1, shall be packed in a shipping container constructed, closed, and strapped to conform to overseas type, style A, type 3 load of Federal Specification PPP–B–601. Each shipping container shall be provided with skids, load bearing members, and additional cleats to support the dishwasher. Not less than two nominal 2- by 4-inch wood skids shall be attached with nails driven from inside of the container and positioned flatwise across the bottom of the container, flush with the outside edges of the container. A staggered nailing pattern shall be used and the nails shall be clinched a minimum of ¼ inch. Ends of each skid shall be beveled full depth at a 45-degree angle. Load-bearing members shall be minimum ¾- by 4½-inch lumber, group III or IV wood, attached to the inside or top surface of the bottom plywood panel and at right angles to the skids. Each dishwasher shall be securely

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bolted either to the load-bearing members and skids of the base and through the additional cleat of one face of the shipping container in a manner as to preclude movement or damage while in transit or storage. Motors, transmissions, pumps, and other suspended components shall be protected by a cushioned cradle, or blocking or holddown so as to preclude vibration, movement, or damage while in transit and storage. A minimum one-inch clearance shall be provided between the dishwasher and the inside surfaces of the sides, ends, and top of the container except the side to which the dishwasher is bolted. Components too large to be placed within the dishwasher packaged as specified in 5.1.1.4, shall be secured to the base of the container under the dishwasher in a manner as to preclude movement or damage during transit or storage.

5.2.2 *Level B.* — Each dishwasher, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that the shipping containers shall be constructed to conform to domestic type, style A of Federal Specification PPP–B–591 or PPP–B–601. Alternatively, shipping containers may be used that are constructed to conform to type II of Military Specification MIL–C–11133, or style B–3 of Military Specification MIL–C–11456. Each wire-bound crate shall be provided with skids of nominal 2- by 4-inch lumber extending the full width of the crate applied to the underside of the base at right angles to the battens at each end of the base. The skids shall be secured to the base by nailing through the skids, battens, and slats with not less than two nails at each point of contact with the battens. Nails shall be spaced approximately two inches apart and clinched a minimum of ¼ inch. Dishwashers packed in crates shall be protected by a shroud fabricated of waterproof barrier-material conforming to any class designation for shrouds in accordance with the requirements of Federal Specification UU–P–271. Seams of the shroud shall be sealed with a continuous seam, minimum ¾-inch width of water-resistant adhesive conforming to Military Specification MIL–A–140.

5.2.3 *Level C.* — Dishwashers, preserved and packaged as specified in 5.1, shall be packed in exterior shipping containers in a manner to insure carrier acceptance and safe delivery at destination. Containers shall meet the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—In addition to any special marking required in the contract or order shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

5.3.3 *Handling marking.*—Arrows and the word "UP" shall be stenciled on the upper one-third of each side of the container and "FRAGILE" and "THIS SIDE UP" shall be stenciled on the top of the shipping container in letters not less than one inch in height.

6. NOTES

6.1 Intended use.—Electric domestic dishwashing machines covered by this specification are designed for under-counter installation. These machines shall render eating utensils physically and hygienically clean.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein and should specify the following:

(*a*) Title, number and date of this specification.

(*b*) Type of dishwashing machine required (see 1.2.1).

(*c*) Whether preproduction sample is required (see 3.1).

(*d*) Whether radio interference suppression is required (see 3.6.18).

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(*e*) Whether fungus resistance is required (see 3.6.2.5).

(*f*) Finish if other than that specified (see 3.8).

(*g*) Whether functional or operational testing is required (see 4.5.2 and 4.5.1).

(*h*) Levels of packaging and packing required (see 5.1 and 5.2).

(*i*) That only one type, class, style, and size shipping container should be used for a specific level of shipment (see 5.2.2).

6.3 Radio interference suppression.—When applicable, the contracting officer should arrange for investigation and test of the suppression components and system by the appropriate agency.

6.4 *Normally required by the Department* of the Navy (see 3.6.2.5 and 3.6.18).

6.5 Transportation description.—The transportation description and minimum weight applicable to this commodity are:

Rail:

Dishwashing machines, household.

Carload minimum weight 16,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Dishwashing machines, household.

Motor volume minimum weight 16,000 pounds.

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MILITARY INTEREST:

Army—Q

Navy—Y Md S MC CG

Air Force—MAAMA

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